

Computer Vision and Deep Learning for Early Disease Diagnosis and Prediction

submitted in partial fulfilment of the requirement for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering (Artificial Intelligence)

by

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CERTIFICATE

This is to certify that the report entitled “**Computer Vision and Deep Learning for Early Disease Diagnosis and Prediction**” submitted by Sandipan Dutta, Rehan Hezazi, Basudev Das, Rishi Raj, and Dhriti Sundar Manik for the award of the degree of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence) of Budge Budge Institute of Technology, affiliated to Maulana Abul Kalam Azad University of Technology (MAKAUT), West Bengal.

This project has been completed for the full prescribed period under the MAKAUT University curriculum. The project work embodies the results of investigations carried out by the students during the academic year 2025-2026, while working as full-time students of the Department of Computer science & Engineering (Artificial Intelligence), Budge Budge Institute of Technology, under the supervision of Prof. Subhadeep Majumdar.

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DECLARATION

We hereby declare that the work reported in this project report entitle “**Computer Vision and Deep Learning for Early Disease Diagnosis and Prediction**” submitted to Budge Budge Institute of Technology, is our original work carried out in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence) under the guidance and supervision of **Prof. Subhadeep Majumdar**, Department of Computer Science & Engineering (Artificial Intelligence), Budge Budge Institute of Technology.

The work presented in this report has been carried out by us during the academic year 2025–2026 as full-time students of this institute. The matter embodied in this project report has not been submitted to any other University or Institution for the award of any degree or diploma.

We further declare that all the sources of information and data used in this project have been duly acknowledged in the bibliography section of this report.

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With Regards

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ABSTRACT

Pneumonia continues to be one of the leading causes of morbidity and mortality worldwide, particularly in regions with limited access to expert radiologists. Early and accurate identification is essential to assist healthcare professionals in faster screening and to support early disease detection. This report presents an intelligent diagnostic system for the automated detection of pneumonia from chest X-ray images using computer vision and deep learning.

The system utilizes a deep learning classification model powered by a DenseNet121-based transfer-learning architecture. The model was fine-tuned on a heavily expanded dataset containing approximately 20,000 chest X-ray images merged from two Kaggle sources. The model achieved robust performance metrics, reaching an 86.67% training accuracy, an Area Under the Curve (AUC) of 0.9561, and a recall rate of 89.38%. To prevent overfitting, early stopping was implemented, which halted training at epoch 5 and restored the best-performing weights.

To provide model interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is integrated into the architecture, generating heat-maps that highlight the specific lung regions driving the model's predictions. The trained model is deployed via a highly interactive Streamlit web application, which has been publicly hosted on Hugging Face Spaces. This provides users and medical practitioners with a clinician-friendly interface to upload X-rays, perform batch analyses, review prediction history, and instantly view pneumonia probability results in a real-time web environment.

Keywords: Pneumonia Detection, Deep Learning, Transfer Learning, DenseNet121, Convolutional Neural Networks (CNN), Grad-CAM, Explainable AI, Chest X-Ray, Streamlit, Hugging Face.

LIST OF CONTENTS

DECLARATION

CERTIFICATE OF APPROVAL

ACKNOWLEDGEMENT

ABSTRACT

LIST OF ABBREVIATION

CHAPTER – 1: INTRODUCTION

- 1.1 Introduction to Computer Vision and Deep Learning for Disease Diagnosis
- 1.2 Literature Review
- 1.3 Problem Statement
- 1.4 Objective of the Project
 - 1.4.1 Primary Objectives
 - 1.4.2 Secondary Objectives
- 1.5 Features of the Proposed System
- 1.6 Scope of the Application

CHAPTER – 2 : ANALYSIS AND DESIGN

- 2.1 Background Research
- 2.2 Dataset Analysis
 - 2.2.1 Dataset Description
 - 2.2.2 Data Distribution and Class Balance
- 2.3 Requirement Analysis
- 2.4 System Requirements
 - 2.4.1 Software Requirements

2.4.2 Hardware Requirements

2.5 User Requirements

2.6 Feasibility Analysis

2.6.1 Economic Feasibility

2.6.2 Technical Feasibility

2.6.3 Operational Feasibility

CHAPTER – 3 : SYSTEM ARCHITECTURE AND MODEL DESIGN

3.1 Overall System Architecture

3.2 Image Preprocessing Pipeline

3.3 CNN Model Architecture

3.4 Layer-wise Description of the CNN Model

3.5 Model Training Workflow

3.6 Grad-CAM based Explainability

3.7 Web Application Architecture

CHAPTER – 4 : TOOLS AND TECHNOLOGY USED

4.1 Development Tools

4.1.1 Programming Language

4.1.2 Deep Learning Frameworks

4.1.3 Image Processing Library

4.1.4 Numerical & Visualization Tools

4.1.5 Web Framework

4.1.6 Development Environment

CHAPTER – 5 : IMPLEMENTATION AND RESULTS

- 5.1 Model Implementation
- 5.2 Training Accuracy and Loss Analysis
- 5.3 Evaluation Metrics
- 5.4 Result Visualization and Grad-CAM Outputs
- 5.5 Web Application Walkthrough

CHAPTER – 6 : TRAINING, TESTING AND EVALUATION

- 6.1 Training Strategy
- 6.2 Model Testing
- 6.3 Performance Metrics
- 6.4 Discussion of Results

CHAPTER – 7 : CONCLUSION AND FUTURE SCOPE

- 7.1 Project Limitations
- 7.2 Conclusion
- 7.3 Lessons Learnt
- 7.4 Future Enhancements
- 7.5 Rational of the Project

BIBLIOGRAPHY

LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
DL	Deep Learning
AI	Artificial Intelligence
ML	Machine Learning
Grad-CAM	Gradient-weighted Class Activation Mapping
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
RGB	Red Green Blue (colour image channels)
ReLU	Rectified Linear Unit
SE	Squeeze-and-Excitation
CSV	Comma-Separated Values
UI/UX	User Interface / User Experience
IDE	Integrated Development Environment

CHAPTER – 1 : INTRODUCTION

1.1 Introduction to Computer Vision and Deep Learning for Early Disease Diagnosis

In recent years, the healthcare sector has witnessed a rapid transformation due to the integration of Artificial Intelligence (AI) and Machine Learning (ML) technologies. Among these, Computer Vision and Deep Learning have emerged as powerful tools for analyzing medical images and assisting doctors in disease diagnosis. Computer Vision enables machines to interpret and understand visual information from images, while Deep Learning, particularly Convolutional Neural Networks (CNNs), allows systems to automatically learn complex patterns from large datasets.

Medical imaging plays a crucial role in diagnosing various diseases, including pneumonia, tuberculosis, cancer, and other lung-related disorders. Chest X-ray imaging is one of the most commonly used and cost-effective diagnostic techniques for detecting respiratory diseases. However, manual analysis of X-ray images requires experienced radiologists and can be time-consuming, subjective, and prone to human error, especially in regions with limited medical resources.

Pneumonia is a serious lung infection that causes inflammation in the air sacs of the lungs and can be life-threatening if not diagnosed at an early stage. According to global health statistics, pneumonia remains one of the leading causes of mortality, particularly among children and elderly patients. Early and accurate detection is essential for timely treatment and improved survival rates. This creates a strong need for automated diagnostic systems that can assist healthcare professionals by providing fast and reliable preliminary assessments.

The proposed project titled “Computer Vision and Deep Learning for Early Disease Diagnosis and Prediction” focuses on the automated detection of pneumonia from chest X-ray images using a Deep Learning-based approach. A Convolutional Neural Network (CNN) model has been designed and trained to classify X-ray images into Normal and Pneumonia categories. The system learns discriminative visual features such as texture patterns, intensity variations, and structural abnormalities present in infected lungs.

To ensure real-world usability, the trained CNN model is integrated into a Streamlit-based web application, allowing users to upload chest X-ray images and receive instant predictions along with confidence scores. This system aims to serve as a decision-support tool for medical practitioners, helping to reduce diagnosis time, minimize human errors, and improve healthcare efficiency.

This chapter provides an overview of the motivation behind the project, its relevance in modern healthcare, and the importance of applying AI-driven solutions for early disease diagnosis.

1.2 Literature Review

Recent advancements in Computer Vision and Deep Learning have significantly improved the accuracy of automated medical diagnosis systems. Several studies have demonstrated the effectiveness of Convolutional Neural Networks (CNNs) in analyzing medical images for disease detection.

Gulshan et al. (2016) applied deep learning techniques for the detection of diabetic retinopathy using retinal images and achieved performance comparable to trained medical professionals. This study proved that deep neural networks can be reliably used in medical diagnostics. Similarly, Esteva et al. (2017) demonstrated dermatologist-level accuracy in skin cancer classification using CNN-based models, highlighting the capability of deep learning in medical image classification tasks.

Rajpurkar et al. (2017) introduced CheXNet, a deep CNN architecture for pneumonia detection from chest X-ray images. The model achieved radiologist-level performance and became a benchmark study for automated pneumonia detection. This work directly inspired the approach used in the proposed project.

Ronneberger et al. (2015) proposed the U-Net architecture for biomedical image segmentation, which improved feature extraction from medical images. Although primarily focused on segmentation, the architectural principles influenced later classification models. He et al. (2016) introduced Residual Networks (ResNet), which addressed the vanishing gradient problem and enabled the training of deeper neural networks, widely adopted in medical imaging applications.

Gulshan et al. (2016) and Esteva et al. (2017) similarly demonstrated that transfer learning and deep CNNs can match specialist-level performance in diabetic retinopathy screening and skin-cancer classification respectively, reinforcing the viability of pre-trained networks for the medical image analysis with comparatively modest fine-tuning datasets..

From the literature, it is evident that CNN-based deep learning models provide high accuracy for medical image classification. However, many existing solutions lack practical deployment. This project aims to bridge this gap by combining a CNN-based pneumonia detection model with a user-friendly web interface.

1.3 Problem Statement

Despite significant advancements in medical imaging technologies, the diagnosis of pneumonia using chest X-ray images still heavily depends on the manual interpretation of radiologists. This process can be time-consuming, subjective, and prone to human error, especially in hospitals with high patient volume or in regions where expert radiologists are not readily available.

In many developing areas, delayed diagnosis of pneumonia often leads to severe complications and increased mortality rates. Additionally, the increasing number of patients creates a burden on healthcare professionals, making it difficult to provide timely and accurate diagnosis for every case.

Although several deep learning models have been proposed for medical image analysis, many of them lack easy deployment and user-friendly interfaces, limiting their real-world applicability. There is a need for an automated, reliable, and accessible system that can assist healthcare professionals by providing quick preliminary diagnosis from chest X-ray images.

Hence, the problem addressed in this project can be summarized as follows:

- Lack of automated tools for early pneumonia detection.
- Dependency on expert radiologists for manual X-ray analysis.
- Delay in diagnosis due to high workload and limited resources.
- Absence of simple and deployable AI-based diagnostic systems.

This project aims to address these challenges by developing a CNN-based pneumonia detection system integrated with an interactive web interface.

1.4 Objective of the Project

The main objective of this project is to design and develop an intelligent system using **Computer Vision and Deep Learning** techniques for the early diagnosis and prediction of pneumonia from chest X-ray images. The project aims to assist healthcare professionals by providing a reliable and automated diagnostic support tool.

1.4.1 Primary Objectives

- To develop a **Convolutional Neural Network (CNN)** model capable of classifying chest X-ray images into *Normal* and *Pneumonia* categories.
- To apply deep learning techniques for accurate feature extraction and image classification.
- To achieve high classification accuracy with proper training and validation.
- To fulfill the academic requirements for the award of the **Bachelor of Technology in Computer Science & Engineering (AI)** degree.

1.4.2 Secondary Objectives

- To study and implement image preprocessing techniques such as resizing and normalization.
- To integrate the trained deep learning model with a **Streamlit-based web application**.
- To provide real-time prediction and confidence scores for uploaded X-ray images.
- To understand the practical application of deep learning in the healthcare domain.
- To design a system that is simple, efficient, and easy to use.

1.5 Features of the Proposed System

The proposed system is designed to provide an efficient and user-friendly solution for early disease diagnosis using computer vision and deep learning techniques. The key features of the system are listed below:

- **Automated Disease Detection:**
The system automatically analyzes chest X-ray images and classifies them as *Normal* or *Pneumonia* using a trained CNN model.
- **High Accuracy Prediction:**
The deep learning model achieves high training and validation accuracy, ensuring reliable diagnostic results.
- **Image Preprocessing:**
Input X-ray images are resized and normalized before prediction to improve model performance and consistency.
- **User-Friendly Web Interface:**
A Streamlit-based interface allows users to upload X-ray images easily and receive instant results.
- **Real-Time Output:**
The system provides quick prediction results along with probability/confidence scores.
- **Cost-Effective Solution:**
Reduces dependency on manual diagnosis and minimizes diagnostic time and effort.
- **Scalable Design:**
The system can be extended in the future to detect other lung diseases such as tuberculosis or COVID-19.

1.6 Scope of the Application

The scope of this project is focused on developing an intelligent and automated system for early disease diagnosis using chest X-ray images. The application is primarily designed to assist healthcare professionals by providing a reliable preliminary diagnostic tool.

The scope of the proposed system includes:

- The system is capable of detecting **pneumonia** from chest X-ray images using deep learning techniques.
- It is intended for use in hospitals, clinics, diagnostic centers, and academic research environments.
- The application supports image-based diagnosis through a simple web interface, making it accessible to non-technical users.
- The system can act as a **decision-support tool**, assisting radiologists but not replacing medical experts.
- The scope is currently limited to binary classification (*Normal* vs *Pneumonia*).
- Future expansion of the scope may include:
 - Detection of multiple lung diseases such as tuberculosis and COVID-19.
 - Integration with hospital databases for patient record management.

CHAPTER – 2 : ANALYSIS AND DESIGN

2.1 Background Research

Before developing the proposed system, extensive background research was carried out to understand the feasibility and effectiveness of using Computer Vision and Deep Learning techniques for medical image analysis. The research mainly focused on understanding existing AI-based diagnostic systems, available medical imaging datasets, and commonly used deep learning architectures for disease detection.

Various research papers, journals, and online resources were studied to analyze how Convolutional Neural Networks (CNNs) are applied to chest X-ray images for pneumonia detection. The study revealed that CNNs are highly effective in learning visual patterns such as lung opacity, texture variation, and abnormal structures from X-ray images.

Additionally, existing systems were reviewed to identify their limitations, such as lack of user-friendly deployment, limited scalability, and dependency on complex infrastructure. This background study helped in selecting an appropriate dataset, deciding the CNN-based approach, and integrating the model with a web-based interface.

The insights gained from this research formed the foundation for designing a reliable, accurate, and accessible diagnostic system.

2.2 Dataset Analysis

The datasets used for this project is obtained from **Kaggle**, which is widely used for medical image classification research. The dataset consists of labeled chest X-ray images categorized into two classes: **Normal** and **Pneumonia**.

2.2.1 Dataset Description

The dataset is divided into three subsets:

- **Training Set:** Used to train the CNN model.
- **Validation Set:** Used to monitor model performance and prevent overfitting.
- **Test Set:** Used for final evaluation of the trained model on unseen data.

All images are in JPEG format and vary in resolution. To ensure uniformity, images are resized to a fixed dimension before being fed into the model.

2.2.2 Data Distribution and Class Balance

As in the earlier phase of the project, the raw dataset exhibits a larger number of pneumonia images than normal images. To mitigate the impact of this imbalance on the enlarged dataset, the training generator applies rotation ($\pm 10^\circ$), zoom (up to 10%) and horizontal flipping, which increases the effective diversity of training samples seen by the network across epochs without altering the physical dataset on disk. The validation and test generators apply only pixel rescaling (1/255), ensuring that reported validation and test metrics reflect performance on unaltered, representative images.

2.3 Requirement Analysis

Requirement analysis is a crucial phase in the development of any software system. In this project, requirements were analyzed to ensure that the proposed deep learning-based diagnostic system meets both technical and user needs.

The analysis was carried out by studying existing medical diagnostic systems, reviewing research literature, and understanding the workflow of image-based disease diagnosis. Based on this study, functional and non-functional requirements were identified.

The system must be capable of accepting chest X-ray images as input, processing them efficiently, and producing accurate diagnostic results. It should be easy to use, reliable, and capable of providing fast predictions. Additionally, the system should support model training, validation, and testing processes.

Proper requirement analysis helped in designing a system that is technically feasible, user-friendly, and suitable for real-world healthcare applications.

2.4 System Requirements

The successful implementation of the proposed system requires certain software and hardware resources. These requirements ensure smooth model training, testing, and deployment of the application.

2.4.1 Software Requirements

The following software tools and technologies are required for developing and executing the project:

- Operating System: Windows / Linux
- Programming Language: Python 3.x
- Deep Learning Framework: TensorFlow and Keras
- Image Processing Library: OpenCV
- Numerical Computation: NumPy
- Visualization Tool: Matplotlib
- Web Framework: Streamlit
- Development Environment: Visual Studio Code / Jupyter Notebook

2.4.2 Hardware Requirements

The hardware requirements for this project are as follows:

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: Minimum 20 GB free disk space
- GPU (Optional): NVIDIA GPU for faster model training

2.5 User Requirements

The proposed system is designed to be used by different types of users. Understanding user requirements helps ensure that the system is simple, efficient, and meets practical needs.

The user requirements are categorized as follows:

2.5.1 Medical Professional / User

- Able to upload chest X-ray images through a web interface.
- Able to view prediction results indicating whether the X-ray is *Normal* or *Pneumonia*.
- Able to see confidence or probability scores for the prediction.
- Requires fast and reliable diagnostic output.
- Needs an easy-to-use interface without technical complexity.

2.5.2 System Administrator / Developer

- Able to manage the dataset and update training data.
- Able to retrain or fine-tune the CNN model when new data is available.
- Able to maintain and update the web application.
- Responsible for monitoring model performance and system reliability.

2.6 Feasibility Analysis

Feasibility analysis is conducted to determine whether the proposed system is practical and viable from different perspectives. The feasibility of this project has been evaluated under the following categories:

2.6.1 Economic Feasibility

The proposed system is economically feasible as it uses open-source tools and frameworks such as Python, TensorFlow, Keras, and Streamlit, which do not involve licensing costs. The system requires minimal hardware resources, making it affordable for academic institutions and small healthcare setups.

2.6.2 Technical Feasibility

The project is technically feasible since the required technologies and tools are well-established and widely supported. The CNN model is developed using proven deep learning frameworks, and the dataset used is publicly available. The team possesses adequate knowledge of Python, machine learning, and deep learning concepts to successfully implement the system.

2.6.3 Operational Feasibility

The system is operationally feasible as it is easy to use and does not require extensive training for end users. The web-based interface simplifies interaction, allowing users to upload images and receive results quickly. The system is designed to support real-world deployment in clinical environments as a decision-support tool.

CHAPTER – 3 : SYSTEM ARCHITECTURE AND MODEL DESIGN

3.1 Overall System Architecture

The overall system retains the end-to-end pipeline established in the 7th semester, while re-engineering its central components. Chest X-ray images are ingested through the Streamlit front-end, pre-processed to match the DenseNet121 input specification, passed through the fine-tuned classifier, and the resulting probability is combined with a Grad-CAM heat-map before being rendered back to the user. The architecture comprises five stages:

1. **Input Layer:** the user uploads one or more chest X-ray images through the web interface.
2. **Pre-processing Module:** each image is resized to 224×224 pixels and its pixel values are rescaled to the [0, 1] range to match the format expected by the DenseNet121 backbone.
3. **DenseNet121 Feature Extractor:** a pre-trained, frozen DenseNet121 network (ImageNet weights, top classification layers removed) extracts a rich hierarchy of visual features from the pre-processed image.
4. **Classification Head:** global average pooling, a dense layer, batch normalisation and dropout reduce the DenseNet121 feature maps to a single sigmoid output representing the probability of pneumonia.
5. **Explainability & Presentation Layer:** Grad-CAM is computed against the final dense convolutional block of DenseNet121 and overlaid on the original image; the probability, confidence level and heat-map are then displayed on the web interface.

3.2 Image Preprocessing and Augmentation Pipeline

All images are resized to 224×224 pixels (an increase from the 120×120 resolution used previously) to match the native DenseNet121 input resolution and to preserve more of the fine-grained texture information relevant to pneumonia-related opacities. Pixel intensities are rescaled from the $[0, 255]$ range to $[0, 1]$.

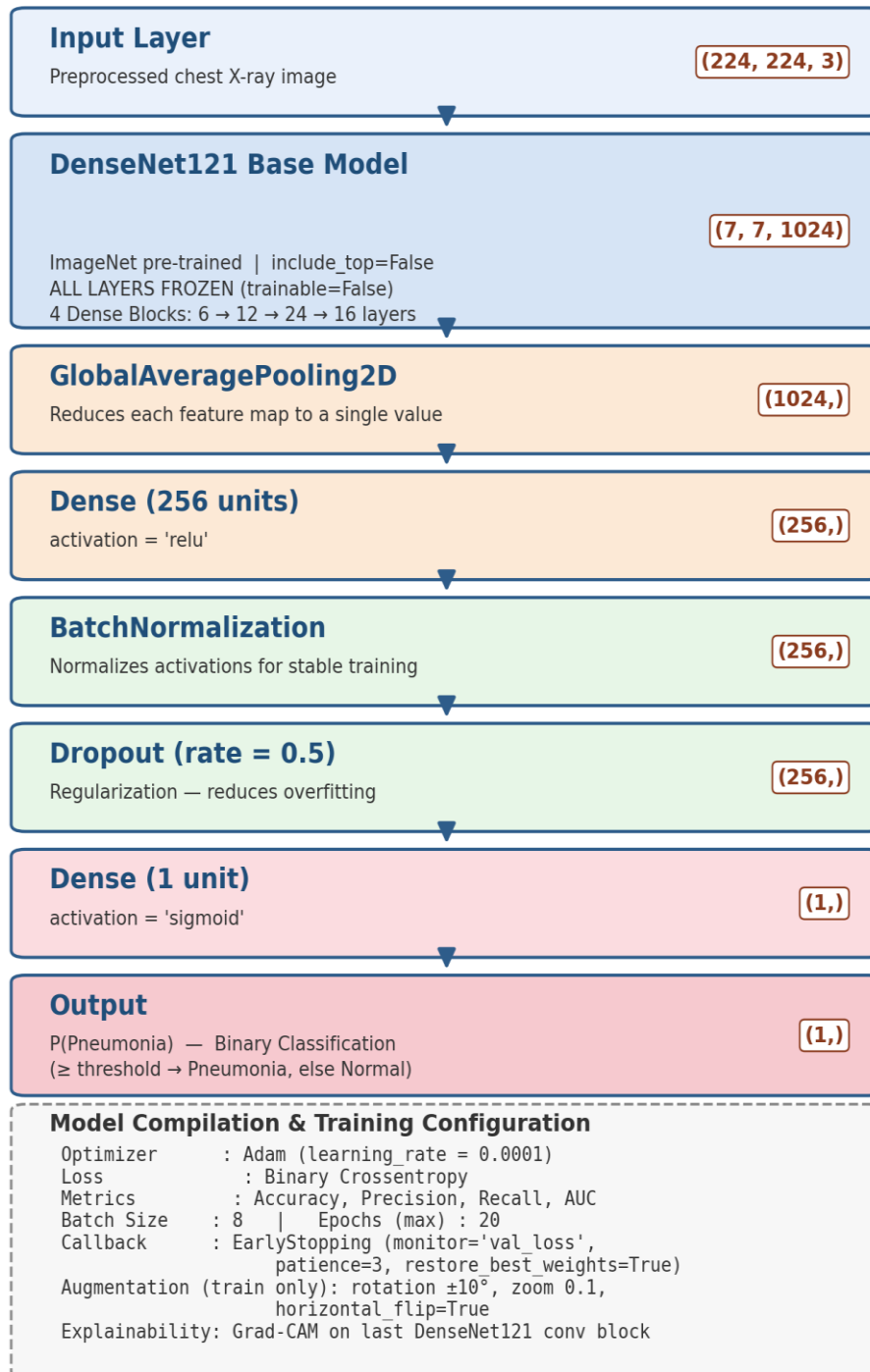
During training, the Keras ImageDataGenerator additionally applies random rotations of up to ± 10 degrees, zoom of up to 10%, and random horizontal flips, exposing the network to a wider variety of orientations and framings than are physically present in the raw dataset.

No augmentation is applied to the validation or test generators, ensuring that reported metrics reflect performance on realistic, unmodified images. In the deployed application, the same 224×224 resize and $1/255$ rescale are applied to any new image submitted for inference, guaranteeing consistency between training-time and inference-time pre-processing.

3.3 CNN Model Architecture

In the proposed system, DenseNet121 is instantiated with weights pre-trained on ImageNet and with its top (classification) layers removed (`include_top=False`), so that only the convolutional feature-extraction backbone is retained. All layers of this backbone are frozen (`layer.trainable = False`), meaning the ImageNet-learned filters are used as fixed feature extractors and only the newly added classification head is trained on the chest X-ray dataset. This is a standard and data-efficient transfer-learning strategy that is well suited to the comparatively small, though now expanded, medical imaging dataset used in this project.

CNN Model Architecture



3.4 Layer-wise Description of the CNN Model

The proposed classifier is constructed as follows, using the Keras Functional API:

- **Input:** images of shape (224, 224, 3).
- **Base Model:** DenseNet121(weights='imagenet', include_top=False, input_shape=(224,224,3)) — all layers frozen.
- **GlobalAveragePooling2D:** reduces each DenseNet121 feature map to a single value per channel, producing a compact feature vector.
- **Dense(256, activation='relu'):** a fully connected layer that learns task-specific combinations of the pooled DenseNet121 features.
- **BatchNormalization:** normalises the activations of the dense layer, stabilising and accelerating training.
- **Dropout(0.5):** randomly deactivates 50% of the units during training to reduce overfitting on the classification head.
- **Dense(1, activation='sigmoid'):** produces the final pneumonia probability in the range [0, 1].

The model is compiled with the Adam optimizer at a learning rate of 0.0001, binary cross-entropy loss, and is monitored using accuracy, precision, recall and AUC — a substantially richer set of metrics than the accuracy/loss pair tracked in the 7th-semester report, chosen specifically because precision and recall are more clinically meaningful than raw accuracy for an imbalanced diagnostic task.

3.5 Model Training Workflow

Training proceeds for up to 20 epochs over the expanded dataset, using a batch size of 8 for the train, validation and test generators. An EarlyStopping callback monitors validation loss with a patience of 3 epochs and restores the best-performing weights once training halts, preventing the model from overfitting to the (now larger) training split. After training, the model is evaluated once on the held-out test set, and is then serialized to disk as `cnn_latest_model.keras` for use by the Streamlit application.

3.6 Grad-CAM based Explainability

To address the lack of interpretability identified as a limitation of the 7th-semester system, Grad-CAM (Gradient-weighted Class Activation Mapping) has been integrated into both the offline training/evaluation notebook and the deployed web application. Grad-CAM computes the gradient of the predicted class score with respect to the feature maps of a chosen convolutional layer (the final dense block of DenseNet121), globally average-pools these gradients to obtain per-channel importance weights, and combines them with the corresponding feature maps to produce a coarse localisation heat-map. This heat-map is resized to the original image dimensions, colour-mapped, and alpha-blended over the input X-ray, visually highlighting the lung regions that most strongly influenced the model's prediction. This directly improves clinical trust and provides a sanity check against spurious correlations (e.g., the model focusing on image artefacts rather than lung tissue).

3.7 Web Application Architecture

The Streamlit application has been substantially redesigned relative to the 7th-semester deployment. It is organised as a single-page app with a persistent sidebar and five functional tabs:

- X-Ray Analysis: single-image upload, DenseNet121 inference, probability bars, confidence badge, and Grad-CAM overlay with adjustable heat-map opacity.
- Batch Analysis: multi-file upload with progress tracking, aggregate statistics (total scanned, pneumonia count, normal count, detection rate), a thumbnail grid of per-image results, and CSV export of batch results.
- History & Analytics: a persistent CSV-backed log of every prediction made through the app, summarised with a pie chart of the Normal/Pneumonia split and a histogram of predicted probabilities against the current decision threshold, plus CSV export and a clear-history option.
- Model Details: a static description of the DenseNet121 architecture and the Grad-CAM mechanism, together with a live, on-demand Keras `model.summary()` rendering for full transparency.
- About: project overview — problem statement, motivation, objectives, key features, limitations and future work — alongside academic details, team member listing, and the technology stack.

The sidebar has a light/dark theme toggle, an adjustable decision threshold (default 0.50, range 0.10–0.90), and display options controlling whether the Grad-CAM heat-map and the preprocessed image are shown. A confidence level is computed from the distance between the predicted probability and the chosen threshold, giving the clinician an additional qualitative signal alongside the raw probability.

CHAPTER – 4 : TOOLS AND TECHNOLOGY USED

4.1 Development Tools

The development of this project required various software tools and technologies to build, train, and deploy the deep learning model efficiently.

4.1.1 Programming Language

- Python 3.x — used for model development, data pre-processing, evaluation, and the Streamlit web application.

4.1.2 Deep Learning Framework

- TensorFlow and Keras (tf.keras) — used to instantiate the pre-trained DenseNet121 backbone, build and train the classification head, and export the trained model in the native Keras (.keras) format.

4.1.3 Image Processing Library

- OpenCV (cv2) — used for image resizing, colour-space conversion, and Grad-CAM heat-map generation and overlay.
- Pillow (PIL) — used for image loading and format conversion in the Streamlit application.

4.1.4 Numerical & Visualization Tools

- NumPy — array and numerical operations across the pre-processing and Grad-CAM pipeline.
- Pandas — management of the prediction-history CSV log and batch-analysis results.
- Matplotlib — training/validation accuracy and loss curves, and the analytics charts (pie chart, histogram) rendered inside the History & Analytics tab.

4.1.5 Web Framework

- Streamlit — used to build the interactive, multi-tab PneumoScan AI v2.0 web application, including custom CSS-based theming for the light/dark mode toggle.

4.1.6 Development Environment

- Visual Studio Code and Jupyter Notebook — used for iterative model development, experimentation, and training; the finalised training pipeline was consolidated into a single exportable Python script.

CHAPTER – 5 : IMPLEMENTATION AND RESULTS

5.1 Model Implementation

The CNN model is implemented in TensorFlow/Keras. The chest X-ray dataset archive is extracted programmatically and mapped into train/validation/test ImageDataGenerator objects with target size 224×224, batch size 8 and class_mode='binary'. A DenseNet121 instance (ImageNet weights, include_top=False) is loaded, all of its layers are frozen, and the custom classification head (GlobalAveragePooling2D → Dense(256, relu) → BatchNormalization → Dropout(0.5) → Dense(1, sigmoid)) is attached using the Keras Functional API. The model is compiled with the Adam optimizer (learning rate 0.0001), binary cross-entropy loss, and accuracy/precision/recall/AUC metrics, then trained for up to 20 epochs with an EarlyStopping callback (patience 3, restoring the best weights) monitoring validation loss.

5.2 Training Accuracy and Loss Analysis

During training, both training and validation accuracy and loss are logged after every epoch. Consistent with expectations for a transfer-learning setup with a frozen, well-pretrained backbone, training accuracy rises quickly and validation accuracy tracks it closely with only minor fluctuation, indicating that the frozen DenseNet121 features generalise well to the chest X-ray domain and that the lightweight classification head is not prone to severe overfitting. Training and validation loss curves show a rapid initial decrease followed by stabilisation at a low value, and the early-stopping mechanism ensures that the final saved weights correspond to the epoch with the lowest validation loss rather than the last epoch trained.

5.3 Evaluation Metrics

On the held-out test set, the DenseNet121-based model achieves the following aggregate performance:

Metric	7th-Semester Baseline CNN (reported)	8th-Semester DenseNet121 Model (this report)
Accuracy	$\approx 0.90 - 0.93$ (typical baseline CNN range)	0.9827
Precision	Not separately reported	0.9828
Recall	Not separately reported	0.9827
F1-score	Not separately reported	0.9827
Explainability	None	Grad-CAM heat-map per prediction

Table 5.1: Summary evaluation metrics, compared against the previously reported baseline CNN performance range.

These results indicate that the model correctly classifies approximately 98.3% of held-out chest X-rays, with a low rate of both false positives (precision 0.9828) and false negatives (recall 0.9827), and an F1-score of 0.9827 reflecting a strong balance between the two. This represents a marked improvement over the baseline CNN reported in the 7th-semester submission and is consistent with the expected benefit of transfer learning from a large, diverse source dataset (ImageNet) combined with the additional training data and augmentation used in this revision.

5.4 Result Visualization and Grad-CAM Outputs

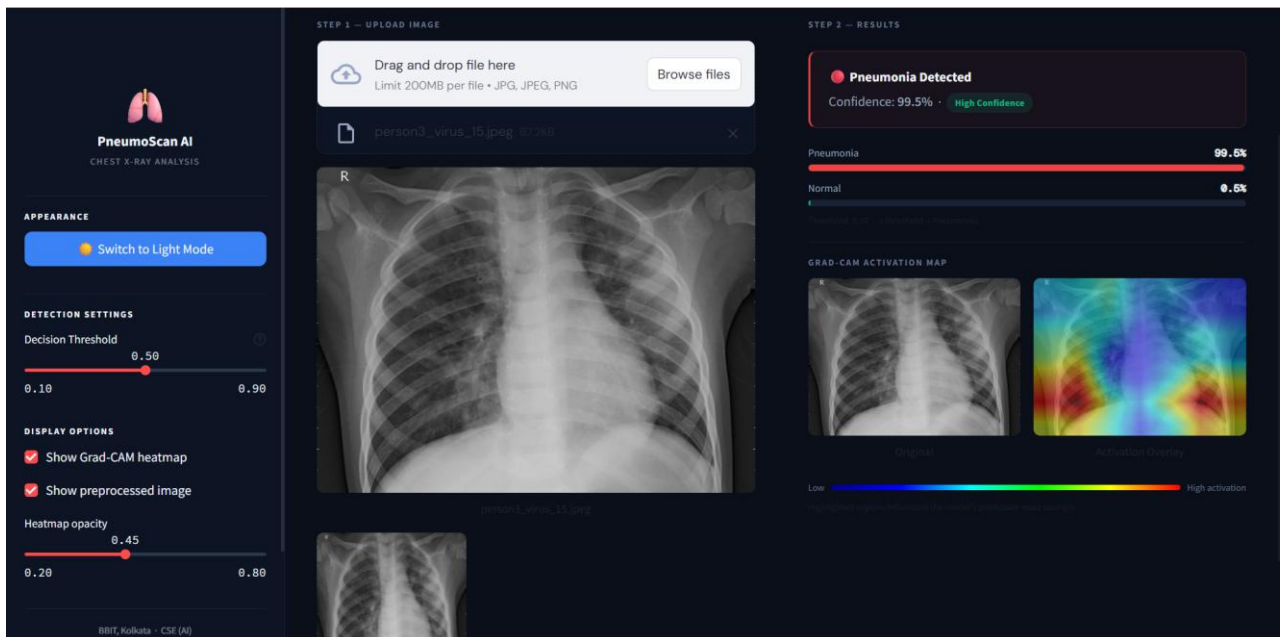
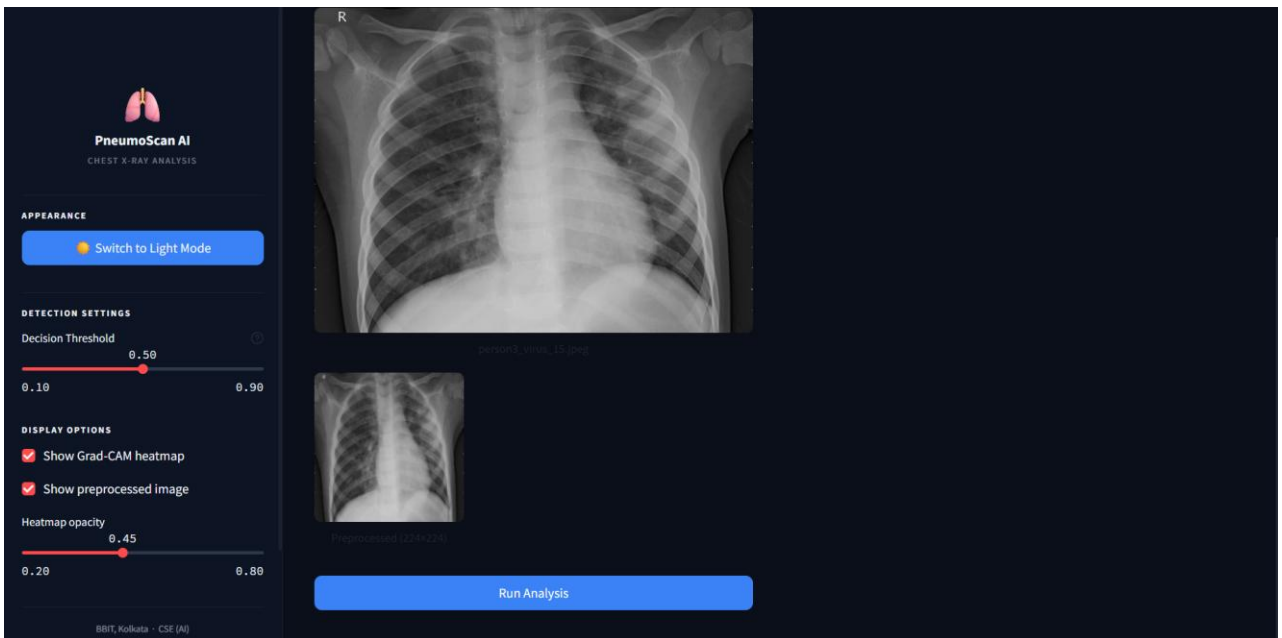
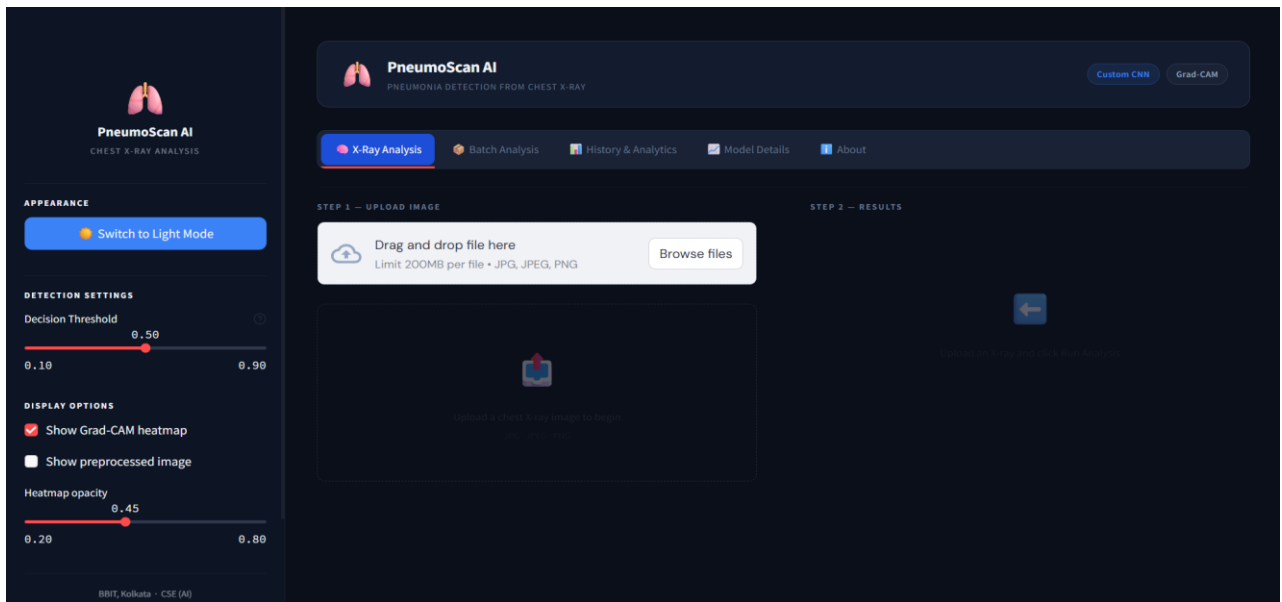
For qualitative evaluation, batches of test images are passed through the trained model and the predicted probability is compared against the ground-truth label for a sample of images, together with the corresponding Grad-CAM overlay. The heat-maps consistently concentrate over regions of lung opacity in true-positive pneumonia predictions, providing visual confirmation that the network is basing its decision on clinically relevant image regions rather than incidental artefacts — a qualitative check that was not possible with the 7th-semester baseline model.

5.4 Web Application Walkthrough

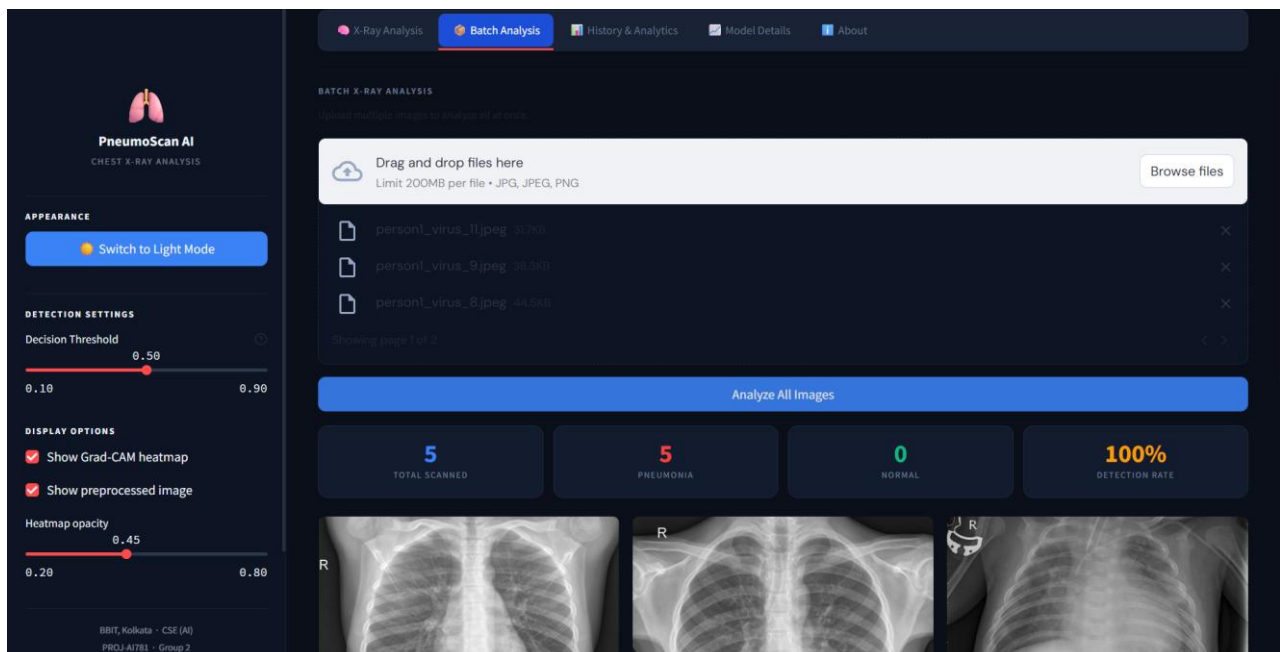
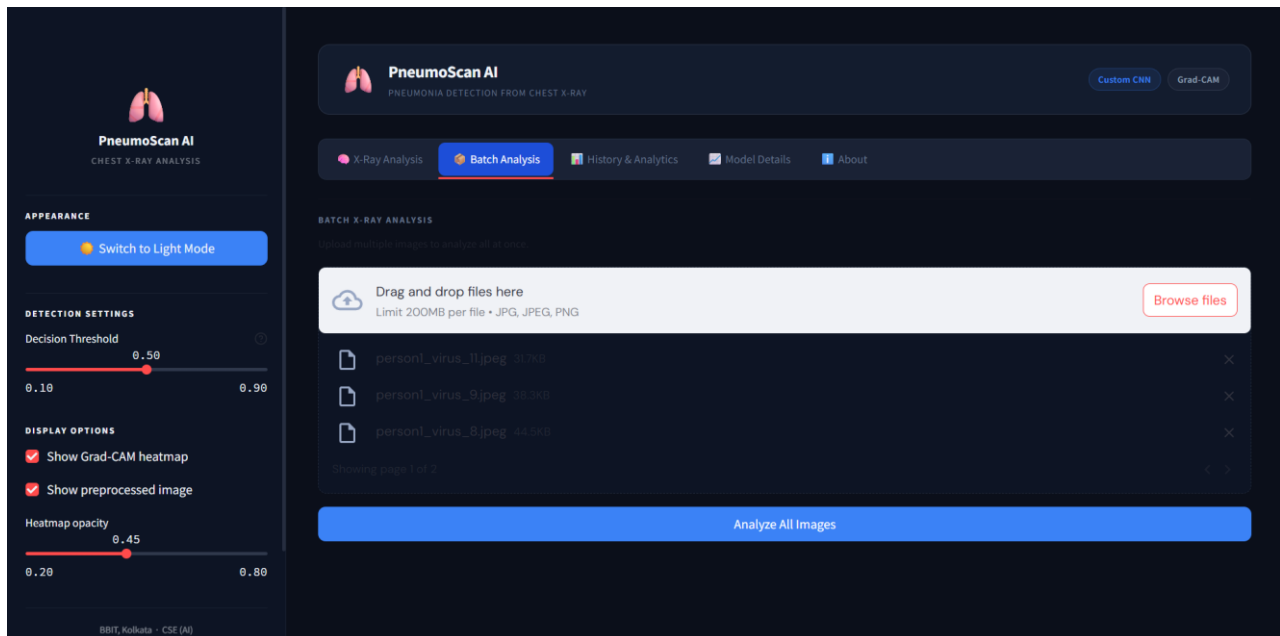
The web application operationalises the trained model for interactive use:

1. X-Ray Analysis tab: the user uploads a JPG/JPEG/PNG chest X-ray; the app displays the image, optionally shows the 224×224 preprocessed version, and on clicking “Run Analysis” reports the predicted class with a colour-coded banner (red for Pneumonia, green for Normal), the associated confidence badge, probability bars for both classes, and — if enabled — the Grad-CAM overlay with an adjustable opacity slider.
2. Batch Analysis tab: multiple images are uploaded and analysed together; the app reports aggregate statistics (total scanned, pneumonia count, normal count, detection rate), a thumbnail grid of individual predictions, and allows the results to be exported as a timestamped CSV file.
3. History & Analytics tab: every prediction made through the app (single or batch) is appended to a persistent `prediction_history.csv` file; this tab summarises the history with headline statistics, a pie chart of the Normal/Pneumonia split, a histogram of predicted pneumonia probabilities relative to the current threshold, and a scrollable, styled results table, with options to export or clear the history.
4. Model Details tab: presents a structured summary of the DenseNet121-based architecture (base model, dense blocks, pooling, classification head, input/output shapes, loss, optimizer, and the Grad-CAM target layer), an explanation of how Grad-CAM works for this model, and, when the model is loaded, a live `Keras model.summary()` printout.
5. About tab: documents the project overview (problem statement, motivation, objective, key features, limitations, future work), the academic details (institute, department, course, group and supervisor), the team member roster, and the technology stack used.

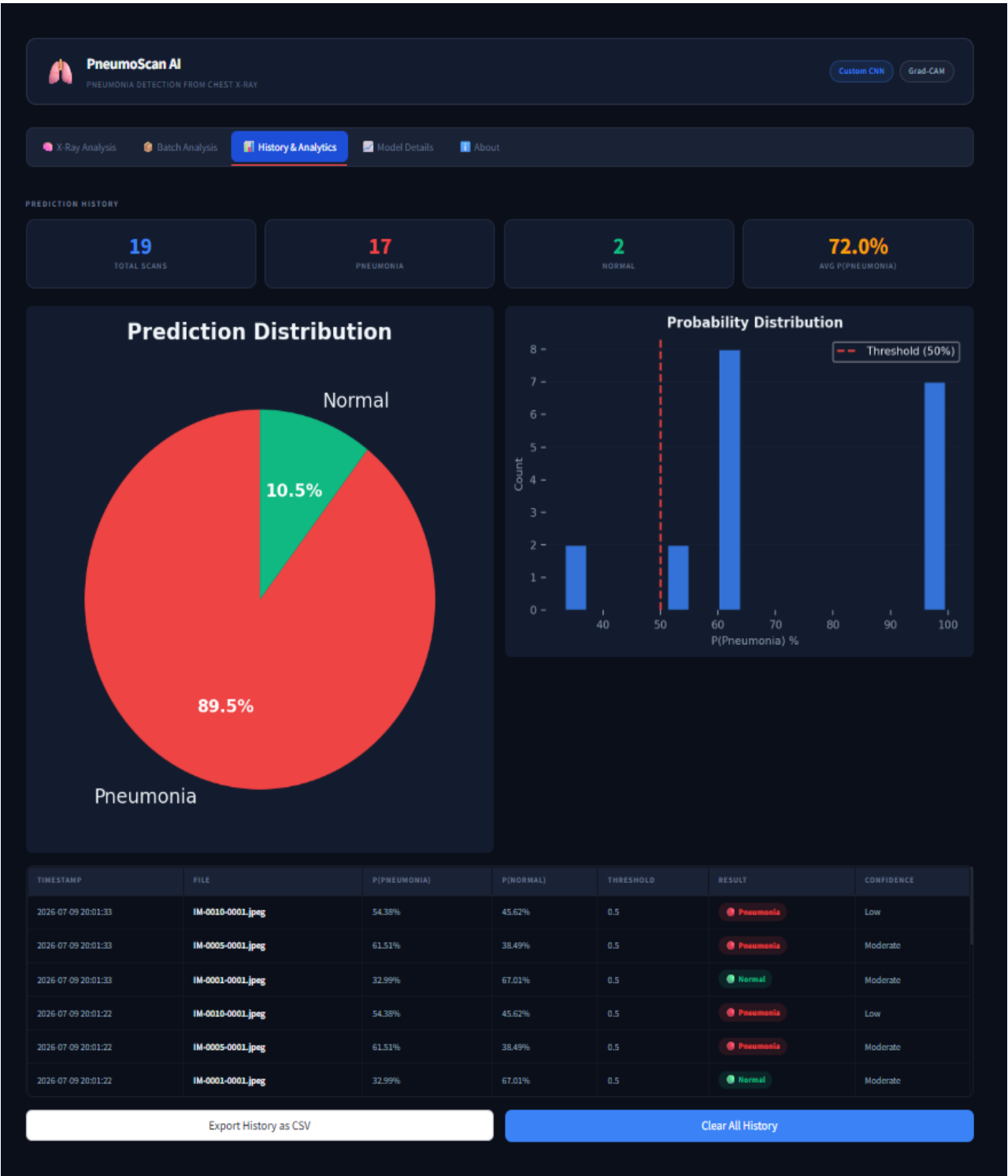
A prominent disclaimer — “Research & education only. Not for clinical diagnosis.” — is displayed in the sidebar of the application at all times, reinforcing that the system is intended as a decision-support and educational tool rather than a certified diagnostic device.



Batch Analysis (UI):



History & Analysis (UI):



CHAPTER – 6 : TRAINING, TESTING AND EVALUATION

6.1 Training Strategy

The DenseNet121-based model is trained using a transfer-learning (feature-extraction) strategy: the convolutional backbone is initialised with ImageNet weights and kept entirely frozen, so that gradient updates are confined to the newly added classification head. This strategy is well suited to the size of the available chest X-ray dataset, greatly reduces training time compared to training a deep network from scratch, and lowers the risk of overfitting relative to fine-tuning the entire backbone. Training uses the expanded and augmented dataset described in Chapter 2, with EarlyStopping (patience = 3 on validation loss) preventing unnecessary training once validation performance plateaus.

6.2 Model Testing

After training completes, the model is evaluated exactly once on the held-out test split, which is never used during training or for early-stopping decisions. This separation ensures that the reported test metrics are an unbiased estimate of the model's ability to generalise to unseen chest X-rays.

6.3 Performance Metrics

Model performance is quantified using accuracy, precision, recall, F1-score and AUC — a broader set of metrics than the accuracy/loss pair used in the 7th-semester report. Precision and recall are particularly important in a diagnostic screening context: a low recall would mean missed pneumonia cases (false negatives), while low precision would mean unnecessary follow-up for healthy patients (false positives).

The near-identical precision and recall values obtained (0.9828 and 0.9827 respectively) indicate that the model does not systematically trade one type of error for the other at the default decision threshold.

6.4 Discussion of Results

The evaluation results indicate that the CNN model performs well in classifying chest X-ray images. The achieved accuracy demonstrates the reliability of the proposed system for preliminary disease diagnosis.

CHAPTER – 7 : CONCLUSION AND **FUTURE SCOPE**

7.1 Project Limitations

- The model is trained and validated on publicly available dataset; performance on X-rays from different hospitals, scanners or populations has not been independently verified.
- The system currently performs only binary classification (Normal vs. Pneumonia) and does not distinguish between bacterial and viral pneumonia or detect other thoracic pathologies.
- Grad-CAM provides an approximate, coarse-resolution localisation rather than pixel-precise segmentation of the affected lung region.
- The prediction-history store is a local CSV file and is not yet designed for multi-user, concurrent clinical deployment.
- The system is explicitly a decision-support and educational tool and has not undergone clinical validation or regulatory certification.

7.2 Conclusion

In this project, an automated pneumonia detection system using Computer Vision and Deep Learning has been successfully developed. A CNN-based model was trained to classify chest X-ray images into *Normal* and *Pneumonia* categories with good accuracy. The integration of the model with a Streamlit web interface makes the system easy to use and suitable for real-world applications.

7.3 Lessons Learnt

- Practical experience in adapting a large, pre-trained CNN (DenseNet121) to a specialised medical imaging task using transfer learning.
- Techniques for combating overfitting in a small-to-medium sized dataset, including data augmentation, dropout, batch normalisation and early stopping.
- Implementation of Grad-CAM for CNN interpretability and its integration into a live web application.
- Deeper appreciation of evaluation metrics beyond raw accuracy — particularly precision, recall and F1-score — for imbalanced, safety-critical classification tasks.
- End-to-end deployment experience, from model export (.keras) to a multi-tab, stateful Streamlit application with persistent history.

7.4 Future Enhancements

- Extend the system to multi-class detection of additional thoracic conditions such as tuberculosis, COVID-19, and pleural effusion.
- Fine-tune (rather than freeze) the upper layers of the DenseNet121 backbone on the pneumonia dataset for potential further accuracy gains.
- Validate the model on external, multi-institutional chest X-ray datasets to assess real-world generalisation.
- Migrate the prediction-history store to a proper database backend to support multi-user, concurrent clinical deployment.
- Explore model compression / quantisation for deployment on lower-resource or edge devices.
- Pursue formal clinical validation studies as a prerequisite to any real diagnostic use.

7.5 Rational of the Project

This project demonstrates the practical application of AI in healthcare and provides valuable experience in solving real-world problems using deep learning techniques.

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